Kunal Pai

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Education

Ph.D., Computer Science, University of California, Los Angeles, *Semiconductor Hub Fellow* Enrolled: September 2026

M.S., Computer Science, University of California, Davis (GPA: **4.0/4.0**) June 2026

Relevant Skills

Languages: Python, C++, C, Rust, JavaScript, Java

ML/AI: LLMs, Multi-Agent Systems, AI Agent Security, PyTorch, TensorFlow, Hugging Face, scikit-learn, Ollama

Web/Data: React, Next.js, Django, Flask, MongoDB, pandas, NumPy

Tools: Git, Docker, Unix/Linux, Jupyter, LaTeX, gem5, RISC-V, LLVM, Clang

Research & Professional Experience

Graduate Student Researcher, *SEAL Lab @ UCLA* September 2026 – Present

- Uncovered 22 divergences across 10 functions in Google's Rust port of giflib, yielding 6 upstream PRs (2 merged)
- Designing boundary-directed differential fuzzing that runs C libraries and their Rust ports on identical, edge-biased inputs
- Building an LLM-agent pipeline that writes fuzzing harnesses, using AST checks to reject restrictive input preconditions

Graduate Student Researcher, *DavSec Lab @ UC Davis* April 2025 – June 2026

- Built an automated pipeline for C-to-Rust transpilation using LLMs, with 5 prompt variations, targeting secure systems migration
- Identified Halstead vocabulary as the strongest metric for predicting translation difficulty
- Validated lightweight semantic augmentations (e.g., filename context) that improved functional accuracy by 5%
- Benchmarked state-of-the-art LLMs across 746 C/C++ programs, achieving 70.2% functional accuracy with best prompt design

Publications (Selected)

VATS: Exploiting Implicit Authority in Error-Path Injection via Systematic Mutation [Spotlight], Patel, H. & Pai, K., *Second Workshop on Agents in the Wild: Safety, Security, and Beyond, ICML 2026*

Toward Reproducible and Standardized Computer Architecture Simulation with gem5, Pai, K., Patel, H., Le, E., et. al., *IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) 2026*

CoDocBench: A Dataset for Code-Documentation Alignment in Software Maintenance, Pai, K., Devanbu, P. & Ahmed, T., *International Conference on Mining Software Repositories (MSR) 2025: Data and Tool Showcase Track*

Calibration and Correctness of Language Models for Code, Spiess, C., Gros, D., Pai, K.S., et. al., *International Conference on Software Engineering (ICSE) 2025*

Automatic semantic augmentation of language model prompts (for code summarization), Ahmed, T., Pai, K.S., Devanbu, P. & Barr, E.T., *International Conference on Software Engineering (ICSE) 2024*

Project Experience

NAAMSE: Neural Adversarial Agent Mutation-based Security Evaluator Nov 2025 – Present

Research Project *Python, LLMs, Evolutionary Algorithms*

- Won 2nd place (Agent Safety) at UC Berkeley RDI AgentBeats Competition; framework infrastructure was subsequently forked by **Mozilla's 0din** team
- Designed and implemented a clustering engine to identify semantic attack vectors in LLM-generated adversarial prompts
- Engineered an attack pipeline that iteratively generates, evaluates, and refines adversarial prompts against target models
- Benchmarked multiple frontier LLMs within the framework to validate attack effectiveness and refine scoring metrics

SuperNOVA: Superconducting Graph Accelerator in gem5 January 2024 – March 2026

Research Project, DArchR Lab @ UC Davis *C++, Python, gem5, Hardware Simulation*

- Modeling a superconducting graph accelerator in gem5, targeting sparse workloads under cryogenic conditions
- Integrating energy and latency estimates from superconducting literature, and RTL simulations, to improve realism
- Mentoring undergraduate researchers on modeling, benchmarking, and research writing; one co-authored a ModSim 2024 poster